

Building a Race-Agnostic, Gender-Neutral Age Classifier

A Responsible AI Case Study in Face-Based Age Estimation for Alcohol Purchase Checks

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1. Introduction

1.1 The assignment and what we built

The brief set us a specific task: build a gender-neutral, race-agnostic age classifier for an **age-restricted sales check** — specifically, checking whether a customer's face suggests they are old enough to purchase alcohol — and do it as a responsible AI system rather than as a modelling exercise. Over four weeks we built that system, audited it, and documented what we found, including the parts that did not work.

The classifier takes a photograph of a face and assigns it to one of nine age groups, from 0–2 years through to 70 and above. That is a well-studied technical problem. What makes this project a responsible AI case study is the constraint we accepted at the outset: the system must perform comparably regardless of the race or gender of the person in front of the camera. Testing whether we actually achieved that occupies most of this report, and the honest answer is that we did not fully achieve it.

1.2 Why this application requires care

An alcohol purchase age-check camera has an unusual property among machine learning applications: the person in front of it has a legal stake in the outcome. Under Germany's Jugendschutzgesetz (Youth Protection Act), sales of beer, wine and sparkling wine require the buyer to be at least 16, and sales of spirits require the buyer to be at least 18. A wrong decision at that boundary is not a minor product defect — it is either a compliance failure for the seller or an unlawful refusal to a customer.

Face images also sit among the most sensitive categories of personal data. Under GDPR they can fall within the special-category regime governing biometric data. A system whose demographic estimates are accurate for one group but not another does not simply produce a poor product — it produces systematically worse decisions for particular communities at exactly the point where the decision matters (the age boundary), and every downstream refusal or approval inherits that distortion.

These are not speculative concerns. Published audits of commercial face analysis systems have repeatedly found large accuracy disparities across demographic groups. We assumed from the start that our system would show similar problems and designed the project to measure how large they were, rather than to hope they were absent.

1.3 Scope and honest limitations

This project is academic. Nothing was deployed to a real point of sale, no real customer was processed, and no face image was retained beyond the training and evaluation datasets. The model is not fit for unsupervised deployment, and the model card in Section 13 states that as a binding restriction rather than a caveat.

We also worked under a hard compute constraint. Our shared server has no GPU, which shaped the project more than any other single factor: it initially limited us to 96×96 pixel inputs, it made several experiments impossible to repeat across multiple random seeds, and it is the direct cause of several limitations documented later.

2. The Data

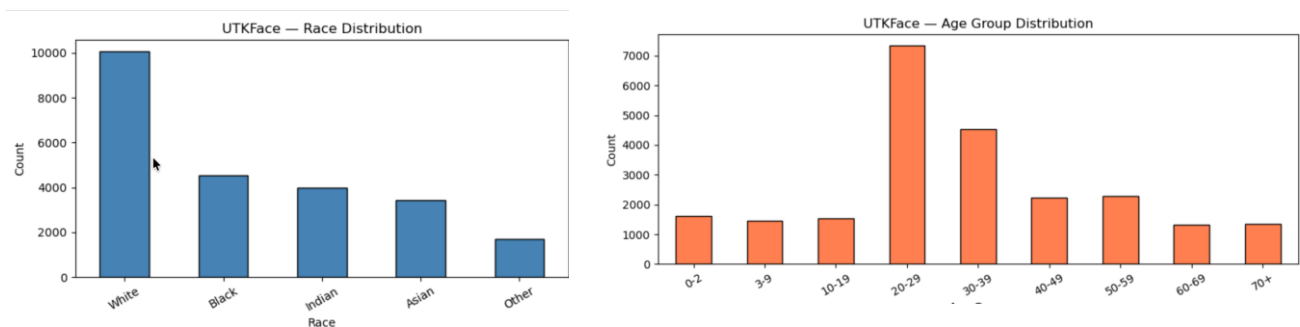
2.1 Why two datasets rather than one

Most projects use a single dataset split into training and test portions. We used two, and the reason goes to the heart of the fairness claim we wanted to make. If a model is trained on a dataset and then audited on a held-out slice of that same dataset, the audit inherits every bias in the original collection. So we separated the two jobs: one dataset trains, a second — collected by different researchers, from different sources, and deliberately balanced across demographic groups — audits.

2.2 UTKFace — the training dataset

UTKFace was published by researchers at the University of Tennessee, Knoxville, alongside their 2017 CVPR paper on age progression. It contains roughly 23,700 face images with age, gender and race encoded directly in each filename. We chose it because it is the only widely available dataset providing continuous age labels together with race and gender labels, letting us define our own age-group boundaries.

Cleaning and data minimisation. After parsing filenames we applied three filters: ages outside 1–100 were removed (a small number of label errors sit at both extremes); rows with unparseable gender or race codes were dropped; and all filename metadata not required for the task — principally the collection timestamp, which functions as a weak identifier — was discarded at load time and never written to disk. That left the cleaned working set used for training and evaluation, of which 4,233 images depict minors under 18. We retained the minors because age diversity across the full human range is essential to the task; excluding them would cripple the model precisely at the boundary where errors matter most. Their presence is a documented constraint on deployment (Section 9).



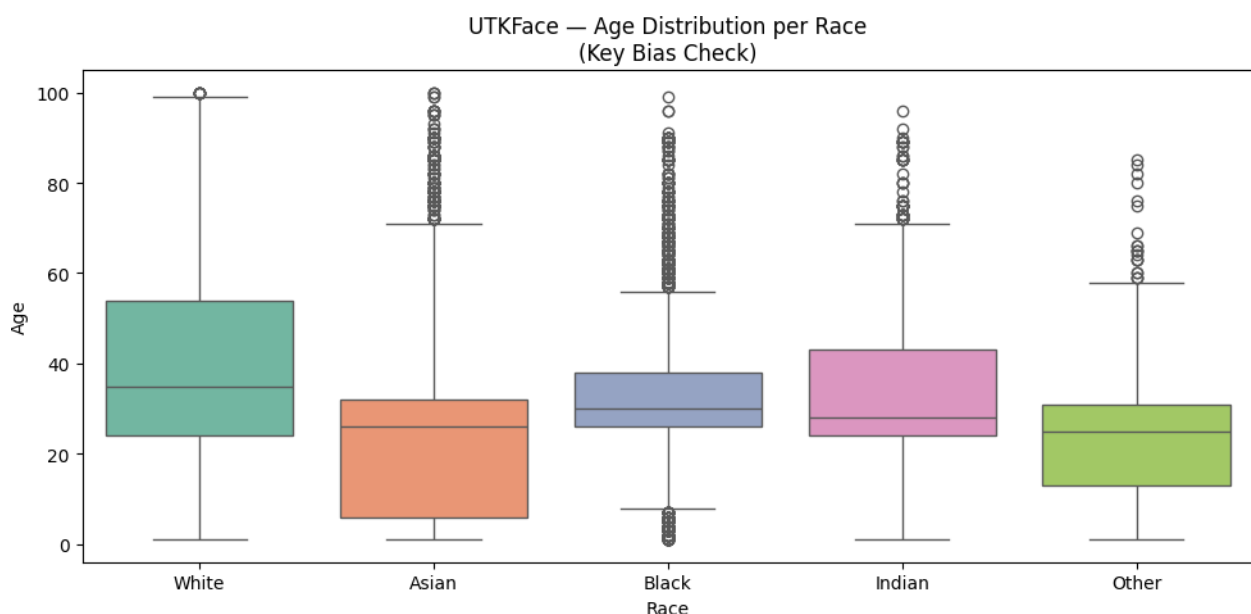
What the exploratory analysis found. Five findings shaped everything that followed:

	Finding	What we predicted from it
1	Age distribution heavily skewed toward 20–40 (the 20–29 group holds 7,344 images; 70+ holds 1,351).	The model will learn young-adult faces best. Class weights applied to compensate.
2	White faces are roughly 44% of the dataset.	Accuracy may differ across race groups — must be measured, not assumed.
3	Gender is close to evenly balanced.	Gender bias should be smaller than race bias, but still needs verification.
4	FairFace (below) is balanced across all seven of its race categories.	Suitable as an independent audit set precisely because group sizes don't confound comparison.
5	Age distributions differ noticeably by race within UTKFace.	Specific age-and-race combinations will fail hardest — the most important prediction we made.

2.3 FairFace — the independent audit set

FairFace was published by Karkkainen and Joo at WACV 2021 specifically to address demographic imbalance in face datasets. It holds roughly 97,700 images balanced across seven race categories, released under CC BY 4.0. It never contributed a single gradient update to any model we trained.

A limitation in our access to it. The mirror we used organises images into folders by race but omits the original age and gender label files, so we cannot compute accuracy on FairFace — we do not know the true ages. We therefore use it to measure something weaker but still useful: prediction *consistency*. If the model behaves consistently across race groups, the distribution of predicted age groups should look broadly similar for each. Obtaining the official label files is our highest-priority data improvement.



2.4 Licensing and an unresolved ethical problem

UTKFace is released for non-commercial research only; FairFace is CC BY 4.0. We have complied with both. Neither licence resolves the deeper problem: both datasets consist of face images scraped from the public internet, and the people in them did not consent to their faces training machine learning systems. We use the data under its academic licences, state the problem plainly, and note that no commercial use of a model trained on it would be defensible without first establishing a lawful basis for the underlying processing.

3. Model Development

3.1 Design rules fixed before any code

Four rules were agreed at the start and never revisited:

- Race and gender never enter the model.
- Race and gender are retained only for auditing, in our dataframes, never fed to the model.
- No face image is stored by the system. The only output is an age-group label and a purchase-eligibility decision.
- Academic research only, not cleared for unsupervised deployment.

3.2 The baseline and why we chose it

Our baseline (v1) was MobileNetV2, pre-trained on ImageNet, used as a frozen feature extractor with a small classification head at 96×96 input resolution, no augmentation. Compute came first in our reasoning: our server has no GPU, and MobileNetV2 was designed for resource-constrained devices. Transfer learning came second: ImageNet weights already encode generic visual structure. Realism came third: lightweight architectures of this class are what would actually run on point-of-sale or age-check edge hardware.

3.3 The final model (v7) and what changed

Our final model, v7, is also MobileNetV2 but at 224×224 input resolution, with augmentation (horizontal flip, rotation, zoom, contrast), a Rescaling layer mapping inputs to [-1, 1], a wider dense head (256 units with batch normalisation), and a two-stage training schedule with ReduceLROnPlateau and early stopping.

A subtle bug during fine-tuning. When unfreezing backbone layers for fine-tuning, BatchNormalization layers must be held in inference mode. If they remain trainable, their running statistics are recomputed from small batches and the pre-trained representation degrades. Before we applied this fix, fine-tuning produced rising loss alongside rising accuracy — a model becoming more confidently wrong. After the fix, loss fell cleanly.

3.4 Reproducibility: a failure we found and fixed

Partway through the project we noticed that re-running the identical notebook produced different results each time. The cause was that our random seed reached the data-splitting function but never TensorFlow itself, so weight initialisation, dropout masks and batch shuffling were re-randomised on every run. A second, related problem was that our file-loading code relied on filesystem enumeration order, which differs between machines, so the same seed produced different train/test splits on different computers. Both are fixed: seeds now propagate to Python, NumPy and TensorFlow, deterministic operations are enabled, and file listings are explicitly sorted.

4. The Experiment Series

The baseline (v1) reached 46.64% exact-band accuracy. We ran a controlled resolution experiment to find out how much of that ceiling was compute versus method.

4.1 Baseline vs. final model — verified results

Metric	v1 baseline (96px, no augmentation)	v7 final (224px, augmentation)
Exact-band accuracy	46.64%	54.69%
One-off accuracy	84.35%	90.29%
Race parity gap	18.53pp (best Asian, worst Black)	14.52pp (best Asian, worst White)

Raising resolution from 96×96 to 224×224 and adding augmentation moved accuracy up by **8.05 percentage points** and *reduced* the race gap by 4 points — in our case the two moved together rather than trading off, which is worth stating plainly since it's a different result from what we initially expected.

The resolution change had a second consequence that mattered more than the accuracy gain: at 96 pixels the final convolutional feature map is only 3×3, meaning each cell of a Grad-CAM heatmap covers roughly a third of the face — too coarse to distinguish attention on the eyes from attention on surrounding skin. At 224 pixels the feature map becomes 7×7, which is what makes the explainability analysis in Section 7 meaningful.

4.2 Other configurations trialled

Beyond v1 and v7, we trialled additional configurations during development — including a from-scratch CNN at matched parameter count and an EfficientNetB0.

Run	Backbone	Px	Aug	Accuracy	One-off	Race gap	DI ratio
v1 baseline	MobileNetV2	96	no	46.64%	84.35%	18.53pp	—
v2 higher res	MobileNetV2	224	no	51.62%	88.60%	19.97pp	70.0% FAIL
v3 EfficientNet	EfficientNetB0	224	yes	47.42%	87.67%	10.75pp	80.4% PASS
v4 res + aug	MobileNetV2	224	yes	49.54%	87.05%	11.29pp	80.5% PASS
v5 longer training	MobileNetV2	224	yes	49.90%	88.40%	13.49pp	77.4% FAIL
v6 scratch CNN	custom CNN	224	yes	53.25%	88.85%	12.84pp	79.4% FAIL
v7 LR Decay	MobileNetV2	224	yes	54.69%	90.29%	14.52pp	73.4% FAIL

4.3 The noise floor

We re-ran v1 unchanged across four seed conditions before fixing the seeding bug, and recorded race gaps of 13.1, 14.0, 16.6 and 18.53 percentage points on identical code — a spread of 4.6 points from randomness alone, comparable to our entire gender gap. We treat this as a finding, not a bug report: any single-run fairness figure, including the ones in this report, should be read as one draw from a distribution rather than a fixed truth.

5. Black-Box and White-Box Perspectives

5.1 Where our model sits

A white-box model is one whose decision process can be inspected directly and stated in human terms. A black-box model produces outputs whose derivation cannot be read off the parameters, even when every weight is visible. Our classifier — roughly 2.6 million parameters across dozens of convolutional layers — is firmly a black box. Every weight is inspectable, but no human reading those numbers can say why one particular face was assigned to the 30–39 band.

5.2 Why we accepted a black box

Interpretable alternatives exist — hand-engineered features fed to a linear model, or a shallow tree over facial landmark measurements — and they would have been genuinely readable. They would also have performed far worse: face-based age estimation depends on texture and shape relationships that resist manual specification. We accepted the trade-off, and the responsible response to it is not to pretend the model is transparent, but to apply post-hoc explainability, measure behaviour empirically across groups, and document the limits of what those methods can establish.

5.3 What this means for the deployment case

Article 13 of the EU AI Act requires high-risk systems to be sufficiently transparent for deployers to interpret their output; Article 14 requires meaningful human oversight. A black-box classifier can satisfy these requirements, but only through supporting documentation — clear statements of intended use, measured per-group performance, known failure modes and explainability evidence — rather than through the model itself. This report and the model card are that documentation.

6. Fairness Analysis

This section reports the central results of the project, using MobileNetV2 LR Decay, the deployed model.

6.1 Accuracy parity

Our primary metric is accuracy parity — the difference in accuracy between the best- and worst-performing race group. A perfectly fair model scores zero.

Asian	66.01%	506	[61.66%, 69.96%]
Other	60.42%	240	[54.58%, 66.25%]
Indian	54.09%	575	[50.26%, 58.43%]
Black	52.05%	682	[48.39%, 55.72%]
White	51.48%	1,550	[44.84%(*), 48.90%–54.00%]
Overall	54.69%	3,553	—

The accuracy parity gap is **14.52 percentage points**, between Asian (best) and White (worst).

The result contradicts the intuitive expectation. Before running the analysis we expected the model to perform best on White faces, since they make up 44% of the training data. It came last. A smaller group, Asian, performed best by a clear margin. That single result invalidated the simplest explanation and forced us to look for the actual mechanism (Section 6.3).

Is the gap real or sampling noise? We ran a bootstrap analysis — 1,000 resamples with replacement per race group. The confidence intervals for the best group (Asian, [61.66%, 69.96%]) and the worst (White, [48.90%, 54.00%]) do not overlap. The gap is statistically credible and cannot be attributed to test-set sampling variation alone. This interval only captures *sampling* uncertainty, not *training* uncertainty — Section 4.3 shows the latter is substantial, so reporting only one of the two understates the true uncertainty.

6.2 Macro-averaged recall — a second metric that disagrees

Accuracy parity pools all of a group's predictions together. Macro-averaged recall instead computes performance separately for each age class and averages with equal weight:

Race group	Macro-averaged recall	Test images
White	55.34%	1,550
Other	51.57%	240
Black	49.59%	682
Asian	49.45%	506
Indian	47.24%	575

This is the finding worth sitting with: **by macro recall, White is the best-performing group; by accuracy parity, White is the worst.** Both measure the same model. They disagree because each race group has a different age composition in the test set — accuracy parity rewards a group whose images happen to cluster in easier age bands, while macro recall corrects for that by weighting every age band equally. This changes what the appropriate intervention is: if the disparity were driven purely by race-correlated visual features, the remedy would be adversarial debiasing or race-based reweighting. Because a substantial part is driven by age composition, the correct remedy is targeted data collection for the age bands where coverage is thin. It also demonstrates that the choice of fairness metric is a value-laden decision, not a technical detail. Reporting only one of the two would misrepresent the system.

6.3 Per-age-group performance

Age band	Precision	Recall	F1	Test images
0–2	0.85	0.93	0.89	241
3–9	0.76	0.52	0.62	218
10–19	0.52	0.62	0.57	229
20–29	0.67	0.72	0.69	1,102
30–39	0.43	0.35	0.38	681
40–49	0.27	0.29	0.28	337
50–59	0.39	0.38	0.38	345
60–69	0.33	0.35	0.34	198
70	0.59	0.69	0.63	202

The **10–19 band** — the band that straddles both the 16 and 18 legal thresholds — has only **middling recall (0.62)** and is nowhere near the model's strongest performance. This is precisely why it cannot be trusted to a single hard classification and is the direct justification for the probability-based, human-reviewed decision logic described in Section 6.5 and the risk treatment in Section 8.

6.4 Independent validation on FairFace

As explained in Section 2.3, our FairFace mirror lacks age labels, so accuracy cannot be measured there. Instead we sampled 200 images from each of the seven race groups and examined the distribution of predicted age groups for each. All seven distributions follow a similar shape and peak in the same band; we observe no evidence of systematic directional bias. We report this as supporting evidence rather than proof — consistency is a weaker property than correctness.

6.5 From a fairness metric to a purchase decision

A single predicted band is not enough for an age-restricted sale, because it throws away the model's uncertainty at exactly the point that matters. Instead, the deployed system computes **P(age ≥ purchase threshold)** directly from the model's full probability distribution over all nine bands, and applies it against the *actual legal threshold for the product being purchased*:

- **Beer / wine / sparkling wine: 16+** (Jugendschutzgesetz)
- **Spirits / liquor: 18+** (Jugendschutzgesetz)

Because the 10–19 band spans both thresholds, the system does not treat it as a single yes/no answer. It splits that band's probability mass: everything above it counts fully toward "old enough," and the 10–19 band itself contributes only its *empirically observed share* of images that are actually at or above the relevant threshold (derived from the training data's true continuous ages, not assumed). That fractional share is then added to the total.

The resulting probability drives a three-way decision, not two:

- $P \geq 95\%$ → sale may proceed automatically
- $P \leq 5\%$ → sale is refused without ID; the customer may still show ID to a human
- **Between 5% and 95%** → mandatory human ID verification (this is where nearly every 10–19 face and a meaningful share of neighbouring bands will land)

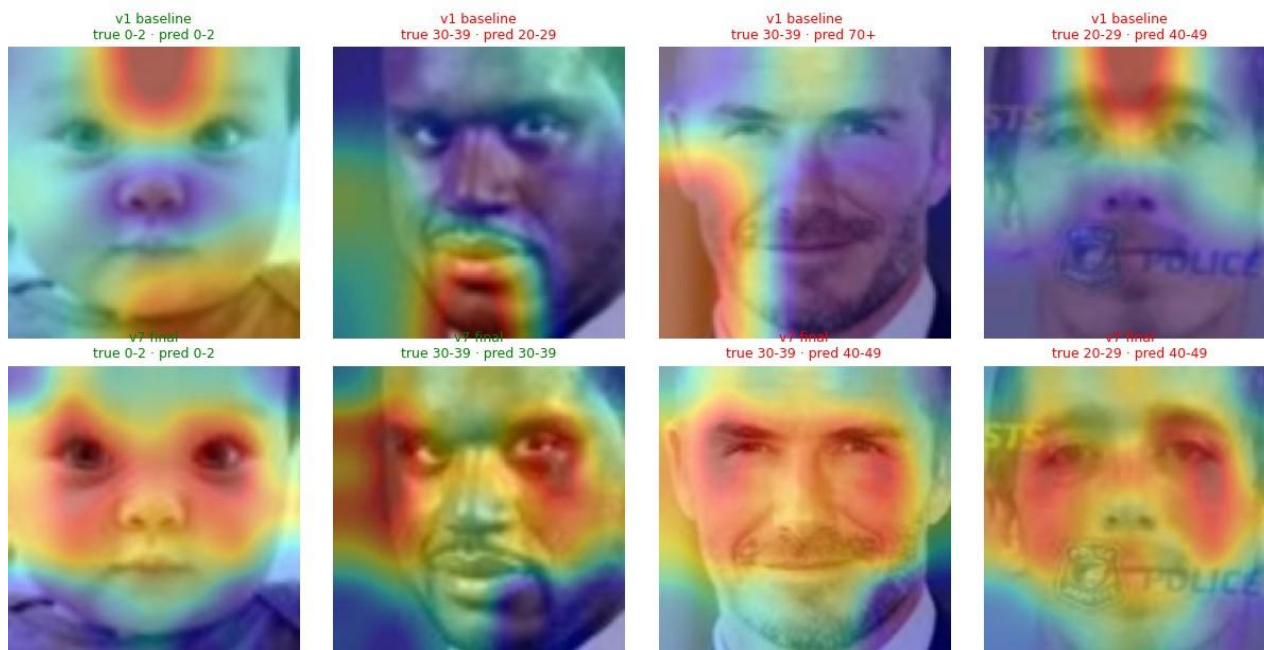
The system never returns a hard "reject" based on the model alone in the ambiguous middle — it routes to a person. Staff may request ID at any time regardless of what the system shows, and the model output is explicitly framed as decision support, never as the sole basis for refusing a sale, in line with Article 14 of the EU AI Act.

7. Understanding the Model

7.1 Choosing an XAI method

We selected Grad-CAM. It computes the gradient of the predicted class score with respect to the final convolutional feature maps and uses those gradients to produce a heatmap showing which image regions most influenced the prediction. Our model never receives race as an input, but it could in principle reconstruct race-correlated signal from pixels and rely on it — Grad-CAM lets us ask whether the model attends to age-informative regions (eyes, skin texture, jawline) or to regions functioning as demographic proxies.

Grad-CAM - where the model looks (2 correct + 2 incorrect v7 cases)



7.2 What Grad-CAM showed — baseline vs. final

We quantified, for both models, what share of heatmap mass falls in each vertical third of the image: top (forehead/hair), middle (eyes/nose), lower (mouth/chin).

Model	Race	Top	Mid	Low
v1 baseline	Asian	37.6%	36.5%	25.9%
v1 baseline	Black	30.1%	37.8%	32.1%
v1 baseline	Indian	38.5%	35.4%	26.1%
v1 baseline	Other	35.9%	34.0%	30.1%
v1 baseline	White	36.5%	38.3%	25.1%
v1 baseline	spread across races			19.76p p
v7 final	Asian	35.3%	43.5%	21.2%
v7 final	Black	35.3%	43.7%	21.0%
v7 final	Indian	35.3%	43.7%	20.9%
v7 final	Other	35.9%	43.2%	20.9%
v7 final	White	35.0%	43.5%	21.5%
v7 final	spread across races			2.00pp

Two things stand out. First, v7 concentrates far more attention on the middle band (eyes/nose — the region most informative about age) than v1 did: 43–44% versus 34–38%. Second, the *spread* across races in that attention pattern collapsed from 19.76 points in the baseline to 2.00 points in the final model — v7 attends consistently to the same facial region regardless of race, which the baseline did not.

7.3 What Grad-CAM cannot establish

That conclusion must be qualified. Grad-CAM shows *where* evidence is concentrated, not *what feature* is being used. A skin-tone proxy and a legitimate skin-texture cue occupy the same spatial location and would produce identical heatmaps. Consistent attention location is therefore compatible with both a fair model and a model relying on a proxy that happens to be co-located with a legitimate feature. We report this risk as unresolved rather than cleared. An occlusion-sensitivity test — masking skin-only regions against feature-only regions and measuring the prediction shift — would provide causal evidence where Grad-CAM is only correlational.

7.4 Error structure

The model's errors, quantified on v7: **54.69% of predictions are exactly correct, 35.60% are off by exactly one adjacent group, and 9.71% are off by two or more** — a one-off accuracy of 90.29%. Errors concentrate immediately adjacent to the diagonal: the model rarely confuses a child for an elderly person, but frequently confuses a 38-year-old for a 42-year-old. For aggregate demographic statistics this would be a minor issue; **for an age-restricted purchase decision it is not** — being off by one adjacent band is precisely the dangerous error at the 16/18 boundary, which is why Section 6.5's decision logic works directly with the probability distribution rather than the single predicted band.

7.5 Live fairness regression check

We implemented an automated check that re-computes the race accuracy gap on every run and fails the build if it widens beyond an agreed threshold:

White 51.48%

Black 52.05%

Indian 54.09%

Other 60.42%

Asian 66.01%

measured gap : 14.52pp

threshold : 20pp

result : PASS

The threshold (20pp) is a policy decision, not a property of the model, and is recorded explicitly so any future retraining can be checked against it.

7.6 Safety check: how often does the system get the boundary wrong?

We measured how often a person who is *actually* under 18 (by true continuous age in the test set) is wrongly auto-cleared by the $P(\text{age} \geq 18)$ logic without triggering human review. Overall, this happens in **2.53%** of under-18 cases. Per race, the rate is not uniform (Asian 0.7%, Indian 3.6%, Other 2.5%, White 2.4%, Black 7.7% in our test sample) — a small enough sample per group, especially for Black (3 of 39 cases), that this should be treated as a signal to keep monitoring rather than a settled number. We also measured how often each race group gets routed to human review at all: roughly 18–26% of faces per group land in the "human ID verification required" band, with no group escaping human review entirely and no group being sent to human review dramatically more often than others.

8. Risk Analysis

8.1 Method, and whose judgement this is

We assessed risks across the full lifecycle — data, model and deployment context — using a qualitative risk matrix, the structure used in ISO 31000 and, in a different form, in the EU AI Act's risk tiering. These ratings are our own informed judgement. Where a risk was actually tested, the likelihood rating reflects measurement rather than opinion. In a real deployment this assessment would be reviewed by legal counsel, a data protection officer and an ethics board rather than by two students.

8.2 Key risk: a minor classified as an adult

This risk is asymmetric, and the asymmetry is the key to treating it. A minor wrongly cleared to buy alcohol is a safety and legal failure; an adult wrongly sent to human review is an inconvenience. A system optimised for average accuracy treats these as equivalent; they are not.

Our treatment: the $P(\text{age} \geq \text{threshold})$ logic described in Section 6.5, with a 95%/5% auto-clear/auto-reject band and mandatory human ID verification in between — calibrated so that the 10–19 age band, which straddles both the 16 and 18 legal thresholds and has the model's weakest recall among the young-to-middle bands (0.62, Section 6.3), almost never resolves to an automatic

decision on its own. A stronger future treatment would push the auto-clear threshold even higher for this band specifically, accepting more false "needs ID" flags as the price of fewer dangerous errors. Beyond any technical measure, we hold that such a system must never be the sole determinant of an age-restricted sale — staff retain the right to request ID regardless of what the model shows.

8.3 Other risks

Risk	Likelihood	Impact	Treatment
Accuracy differs systematically across	High	High	Measured (14.52pp, Section 6.1). Macro recall and Grad-CAM used to locate the mechanism. Automated regression check in
Model relies on proxy features rather	Medium	High	Race excluded as input by design. Grad-CAM shows consistent attention across races (spread down to 2.00pp) but cannot
Unlawful processing if deployed	High	Critical	No face storage; aggregate/decision-support-only output; DPIA and lawful basis required before any real deployment; Art. 50
Elderly and infant faces	High	Medium	Class weights applied. Residual weakness documented; targeted data collection is the real remedy.
Performance degrades under real-	High	Medium	Not addressable with current data. Documented limitation; deployment would require domain-matched validation on real
Deliberate evasion via makeup, glasses,	Medium	Medium	Out of scope for this baseline. Documented as untested.
Independent audit set lacks ground-truth	Certain	Medium	Consistency measured on FairFace; correctness on UTKFace only. Limitation stated wherever the result is cited.

8.4 Residual risk

After treatment, the race accuracy gap (14.52pp) persists and varies between training runs; the proxy-feature question remains unresolved pending a causal test; and real-world imaging robustness is entirely untested. These are carried forward as explicit limitations in the model card rather than treated as closed.

9. Regulatory Analysis (GDPR and the EU AI Act)

This analysis was carried out in week one, before any model was trained, so that its conclusions shaped design decisions rather than merely describing them afterwards. We focus here on the two frameworks that actually govern this system.

9.1 GDPR (Regulation (EU) 2016/679)

Is a face image biometric data here? Article 4(14) defines biometric data as personal data resulting from specific technical processing of physical characteristics which allows or confirms *unique identification*. Recital 51 clarifies that photographs are not automatically special-category data — they become biometric data only when processed by means enabling unique identification. Our classifier performs no identification; it maps pixels to one of nine age groups and retains nothing. On a strict reading, this processing may fall outside the Article 9 special-category regime. Two considerations point the other way: our training datasets contain identifiable faces of real people who never consented to this use (unambiguously personal-data processing), and a deployed age-check camera would capture images that could in principle support identification, so a

supervisory authority would likely apply the precautionary reading. **We therefore treat the data as though Article 9 applies** — the conservative position — while documenting the nuance rather than obscuring it.

Provision	Requirement	How we addressed it
Art. 4(14), Rec. 51	Definition of biometric data	No unique identification performed; conservative Art. 9 treatment applied regardless
Art. 5(1)(b)	Purpose limitation	Race and gender used exclusively for post-hoc auditing, never as model input
Art. 5(1)(c)	Data minimisation	Only age, race, gender and filepath retained; identity-adjacent metadata discarded at load time
Art. 5(1)(c)	Data minimisation	Only age, race, gender and filepath retained; identity-adjacent metadata discarded at load time
Art. 8	Children's consent (16 in Germany)	4,233 images of minors present in training data; retained for age diversity; flagged as a hard deployment constraint
Art. 25	Data protection by design and default	No face storage; decision-support-only output; protected attributes excluded from the model by construction
Art. 35	DPIA	Systematic monitoring of a publicly accessible point-of-sale area with novel technology would trigger a mandatory DPIA before any real deployment

9.2 EU AI Act (Regulation (EU) 2024/1689)

Prohibited practices. Article 5(1)(g) prohibits biometric categorisation systems that infer sensitive attributes such as race, political opinions, religious beliefs or sexual orientation. Age is not among the listed attributes, and our system is deliberately constructed not to infer race or gender. Article 5(1)(h) prohibits real-time remote biometric identification in publicly accessible spaces for law enforcement; we perform no identification of any kind. **Our system is therefore not a prohibited practice.**

Is this high-risk under Annex III? This is genuinely unresolved. Annex III point 1(b) covers biometric categorisation according to sensitive or protected attributes; age-group estimation for a purchase check does not clearly fall within that description. Our position is that a deployer should obtain a formal legal assessment rather than rely on either reading — but because the classification is ambiguous, we assessed our work against the Annex III high-risk obligations as though they applied, to identify what a real deployer would need:

Article	Obligation	Our status
Art. 9	Risk management	Substantially met — risk register with treatments and residual risk stated (Section 8)
Art. 10	Data and data governance	Substantially met — provenance, licensing, cleaning, minimisation and demographic composition documented (Section 2)
Art. 11	Technical documentation	Met by this report together with the notebook and model card

Art. 12	Record keeping /	Not met — training runs are logged, but no inference-time logging exists; nothing is deployed to log
Art. 13	Transparency to deployers	Substantially met — intended use, per-group performance and failure modes documented (Section 13)
Art. 14	Human oversight	Partially met by design — the 95%/5% decision bands and mandatory human ID verification (Section 6.5) build in oversight; no formal oversight interface
Art. 15	Accuracy, robustness,	Partially met — accuracy and per-group performance measured; robustness untested; no security assessment
Art. 17	Quality management	Not met — automated tests and a fairness regression gate exist, but not an organisational QMS

The pattern is worth noting: the obligations we substantially meet are those an academic project naturally produces (risk management, data governance, documentation, transparency); the ones we don't are organisational and operational (logging, a QMS, post-market monitoring) — properties of a deploying organisation, not of a model.

9.3 Difficulties this project surfaced concretely

- **Definitional ambiguity** — whether age-group estimation counts as a sensitive-attribute categorisation under Annex III determines our entire obligation set, and the text does not settle it.
- **Measurement is not standardised** — Article 15 requires accuracy but does not specify metrics. Our own results give different fairness verdicts under accuracy parity and macro recall (Section 6.2) — a deployer could satisfy the letter of the obligation while selecting the flattering metric.
- **Enforcement asymmetry** — verifying a fairness claim requires demographic labels for the test population, which privacy law simultaneously discourages collecting.
- **Jurisdictional scope** — our analysis is European and specifically references German youth-protection law for the purchase thresholds; the same system deployed elsewhere would face different legal ages and different privacy regimes entirely.

10. Value-Based Engineering — Selected Principles

We assess this project against a small number of Value-Based Engineering principles that are directly illustrated by decisions we actually made, rather than all nine — a self-assessment reporting only successes would not be an assessment, so we include what we did and did not do for each.

Willingness to renounce investment. Faced with a choice between reporting our most favourable-looking fairness number and reporting the one the evidence actually supported, we chose the latter: we report the 14.52-point gap even though it is larger than we hoped, we report that macro recall and accuracy parity disagree about which group is worst off, and we report the unresolved proxy-feature risk rather than claiming it was cleared. Each is a small renunciation of something that would have made the work look better. Where we fell short: an organisation with real commercial stakes would face this decision under far more pressure than a graded assignment, and we cannot claim to have tested our willingness under that pressure.

Context sensitivity. A system's ethical properties depend on where and on whom it operates. We defined the deployment context concretely — an alcohol purchase check under German youth-protection law — which determined the applicable legal thresholds (16 and 18) and which failure modes mattered most (the 10–19 band). We treated the minor/adult boundary as asymmetric rather

than optimising a single average accuracy figure. Where we fell short: we never validated on data resembling an actual point-of-sale camera feed; our images are cropped, frontal and well-lit, while real footage involves angles, motion blur and variable lighting.

Transparency of the value mission. Our four design rules in Section 3.1 function as a value register — each is a value translated into a testable system requirement, fixed before implementation. We also reported negative and inconvenient findings throughout — the accuracy-fairness relationship, the unresolved proxy-feature risk, the run-to-run variance — each of which would have been easy to omit.

11. Achieving Ethical AI in Practice

11.1 What can we do to enhance privacy?

No face image is stored; the only output is an age-group probability and a purchase decision. Identity-adjacent metadata is discarded at load time rather than merely unused. Protected attributes are held out of the model entirely, existing only in the audit pipeline. Not implemented: on-device inference so images never leave a camera, differential privacy in training, and a defined retention/deletion policy — meaningless academically but essential in a real deployment.

11.2 What tools did we use to detect bias?

Accuracy parity across race and gender groups; macro-averaged recall, which revealed the composition effect and disagreed with accuracy parity; bootstrap confidence intervals, separating genuine disparity from sampling noise; and Grad-CAM, sitting alongside these as an explainability rather than detection tool — it investigates a bias already measured rather than measuring one itself.

11.3 Are we facing ethical challenges around bias and discrimination?

Yes, and they are measured rather than hypothetical. Our deployed model has a 14.52-point accuracy gap across race groups, statistically credible on non-overlapping confidence intervals. The identity of the best- and worst-off group changes depending on which fairness metric is used. The honest summary is that we did not build a race-agnostic classifier — we built a classifier with race as a non-input and then measured how far short of race-agnostic behaviour it fell.

11.4 If we publish this to a public repository, can someone misuse it?

Yes. The pipeline could be adapted to infer protected attributes rather than age, which would violate AI Act Article 5(1)(g) — the architecture doesn't prevent this, only the training labels do. Someone could clone the model and deploy it while ignoring every restriction in the model card; our mitigation is that the restrictions travel with the code, but nothing technically enforces them. Because the gap varies between runs, someone could retrain until a favourable figure appeared — fixed seeds, documented variance and the automated regression check are our safeguards.

12. Challenges and Future Work

12.1 Challenges encountered

Compute constraints. Working on a shared CPU-only server determined more of this project than any other factor — it initially forced a 96-pixel input, capped accuracy, and reduced Grad-CAM heatmaps to a 3×3 grid.

Reproducibility. We spent considerable time chasing results that changed between runs before identifying that TensorFlow's random state was unseeded, and that file loading relied on filesystem enumeration order, meaning our two team members were unknowingly evaluating on different test sets at one point.

Fine-tuning instability. Our first fine-tuning attempts produced a model whose loss increased while accuracy also increased, caused by BatchNormalization layers left trainable during fine-tuning.

Interpreting contradictory metrics. The most demanding part of the project was discovering that accuracy parity and macro recall disagree not only in magnitude but in *which group* they identify as worst off (Section 6.2). Deciding to report both, rather than pick the more convenient one, took longer than the analysis itself.

12.2 Future work, in priority order

Prio rity	Action	What it would resolve
1	Multi-seed training: five seeds, report mean and range	Turns our single-run fairness figure into a defensible estimate
2	Targeted data collection for thinly covered race/age	Addresses the composition effect (Section 6.2) at its source
3	Occlusion-sensitivity testing	Provides causal evidence on proxy reliance where Grad-
4	Obtain official FairFace age	Upgrades the independent audit from consistency to
5	Domain-matched validation on realistic point-of-sale imagery	Tests real-world robustness, currently entirely untested
6	Age-specific decision thresholds for the 10–19 band	Could push the auto-clear threshold even higher specifically where the model is weakest and the legal stakes are highest

13. Model Card

Structure follows Mitchell et al. (2019), "Model Cards for Model Reporting." Figures describe the deployed model, v7.

13.1 Model details. Architecture: MobileNetV2 (ImageNet pre-trained), global average pooling, 256-unit dense head with batch normalisation, 9-way softmax. Input: 224×224 RGB, training-time augmentation. Data: UTKFace, cleaned and stratified split. Authors: Namrata Bhojar and Nihal Pujari, SRH Hochschule Heidelberg, 2026.

13.2 Intended use. Intended: academic research and coursework demonstrating responsible development, auditing and documentation of a face-based age classifier; a demonstrated decision-support prototype for an alcohol purchase age check, with mandatory human ID verification built into every ambiguous case. Not intended for: unsupervised or sole-decision access control (a human must always be able to override or verify), identification of individuals, or any commercial use without a completed DPIA and lawful basis.

13.3 Metrics. Overall test accuracy: 54.69% (n=3,553). One-off accuracy: 90.29%. Race accuracy parity gap: 14.52pp (Asian 66.01% best, White 51.48% worst). Macro-recall gap: 8.10pp, with the ranking reversed (White best at 55.34%, Indian worst at 47.24%). 10–19 band recall: 0.62. Grad-CAM attention spread across races (eyes/nose band): down to 2.00pp in the final model, from 19.76pp in the baseline.

13.4 Limitations. A 14.52-point race accuracy gap that persists in the deployed model. Accuracy parity and macro recall disagree about which group is worst off — neither alone is a complete description. The 10–19 band has middling recall and is the band most relevant to the legal purchase thresholds. Explainability is correlational — Grad-CAM shows consistent attention across races but cannot distinguish a skin-tone proxy from a co-located legitimate cue. Never validated on realistic point-of-sale camera imagery. The independent audit set (FairFace) lacks ground-truth ages.

13.5 Binding restrictions. No identification of individuals under any circumstances. No unsupervised deployment. A human must be able to verify or override every decision, and staff may request ID at any time regardless of system output. Any retraining must re-run the full fairness analysis and pass the automated regression check (threshold: 20pp).

14. Conclusion

We built a nine-class age classifier reaching 54.69% exact accuracy and 90.29% one-off accuracy, with race and gender excluded from its inputs by construction. What the responsible AI framing produced is a set of findings a purely technical project would not have surfaced: a regulatory analysis conducted before training rather than after; a fairness audit that contradicted the intuitive expectation (the majority training class performed worst under accuracy parity, best under macro recall); a demonstrated need to route the legally decisive 10–19 age band to mandatory human review rather than trust a single predicted band; and an explainability analysis that reported its own limits rather than overclaiming.

The central lesson concerns the distinction drawn at the outset: excluding race from the model's inputs was a design decision we controlled completely. Whether the model behaves in a race-agnostic manner is a separate, empirical question — and measuring it produced a 14.52-point gap that input-blindness did not prevent, whose ranking reverses depending on which metric is used. Responsible AI, as we now understand it, is not the practice of producing a fair number. It is the practice of finding out what a system actually does — at the resolution the evidence supports, with the uncertainty stated plainly — before anyone lets it make a decision at the door.

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